

Problem 1 (§1.1, #16). Write the following set by listing their elements between braces.

$$\{6a + 2b : a, b \in \mathbb{Z}\}$$

Problem 2 (§1.1, #22). Write the following set in set-builder notation.

$$\{3, 6, 11, 18, 27, 38, \dots\}$$

Problem 3 (§1.2, #4). Write out the indicated set by listing their elements between braces.

$$\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\}$$

Problem 4 (§1.2, #14). Sketch the Cartesian product on the x - y plane $\mathbb{R}^2$.

$$[1, 2] \times \{1, 1.5, 2\}$$

Problem 5 (§1.3, #6). List all the subsets of the following set.

$$\{\mathbb{R}, \mathbb{Q}, \mathbb{N}\}$$

Problem 6 (§1.4, #8). Write the following set by listing their elements between braces.

$$\mathcal{P}(\{1, 2\} \times \{3\})$$

Problem 7 (§1.5, #8). Sketch on $\mathbb{R}^2$ the sets

$$X = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \text{ and } Y = \{(x, y) \in \mathbb{R}^2 : -1 \leq y \leq 0\}.$$

On separate drawings, shade in the sets $X \cup Y$, $X \cap Y$, $X - Y$ and $Y - X$.

Problem 8 (§1.6, #4). Sketch the set $X = [-1, 3] \times [0, 2]$ on the plane $\mathbb{R}^2$. On separate drawings, shade in the sets $\bar{X}$ and $\bar{X} \cap ([-2, 4] \times [-1, 3])$.

Recommended Problems:

§1.1: 7-12, 17, 18, 23-25, 40-44

§1.2: 1-4, 12-18

§1.3: 2-6, 9, 10

§1.4: 1-4, 7-9

§1.5: 1-4, 7-8

§1.6: 1-4

§1.7: All problems, if needed.